

EDUCATION

RWTH AACHEN UNIVERSITY

DR.-ING. (PHD)

INSTITUTE FOR DATA SCIENCE IN

MECHANICAL ENGINEERING

Expected Mar 2027 | Germany

Supervisor: Prof. Dr. Sebastian Trimpe

Associated Researcher at RTG UnRAVeL

M.Sc. AUTOMATION ENGINEERING

Dec 2021 | Germany

Grade: 1.1 (1.0 = best)

With distinction

B.Sc. MECHANICAL ENGINEERING

Mar 2019 | Germany

UNIVERSITY OF CAMBRIDGE

ELLIS SUMMER SCHOOL

Jul 2023 | UK

Probabilistic Machine Learning

GYMNASIUM KRONSHAGEN

ABITUR

Jun 2015 | Germany

DPG-Abitur Award (Physics)

LINKS

 Scholar:// [Paul Brunzema](#)
 GitHub:// [brunzema](#)
 LinkedIn:// [paul-brunzema](#)

SKILLS

PROGRAMMING

Python (JAX, PyTorch, NumPy, ...)

Matlab • Shell • Git • ROS2 • $\text{\LaTeX}$

LANGUAGES

German (native)

English (fluent)

AWARDS & HONORS

SEW-EURODRIVE Student Award

2023 | Best Master's thesis

Dean's List | 2021-22

MathWorks Fellowship | 2021

Germany Scholarship | 2021

DPG-Abitur Award | 2015

HOBBIES

Chess • American football • Hiking

CrossFit • Cycling • Board games

My research lies at the intersection of **uncertainty quantification** (Gaussian processes, Bayesian neural networks) and **sequential decision-making** (Bayesian optimization, control, reinforcement learning). I aim to create agents and algorithms that efficiently and optimally act in uncertain and potentially changing environments.

EXPERIENCE

RWTH AACHEN UNIVERSITY | DOCTORAL RESEARCHER

Mar 2022 – Mar 2027 (expected) | Aachen, Germany

- Research on (time-varying) Bayesian optimization and event-triggered learning
- Co-created two edX MOOCs: [Reinforcement Learning](#), which reached over 600 students across 83 countries in 2025, and [Learning-Based Control](#), which reached approximately 350 students across 58 countries in its first run in 2026

META | RESEARCH SCIENTIST INTERN

May 2026 – Sep 2026 | New York, NY, USA

Project: Agentic Bayesian optimization and surrogate-augmented autoresearch

- Introduced *agentic Bayesian optimization*: an LLM agent as the central decision maker in the optimization loop, informed by an uncertainty-aware BO backend
- The approach allows domain knowledge expressed in natural language, code, or documentation to guide optimization toward faster convergence and lower regret

TOYOTA RESEARCH INSTITUTE | RESEARCH SCIENTIST INTERN

Jun 2025 – Sep 2025 | Los Altos, CA, USA

Project: Vision-conditioned dynamics models for learning-based control

- Developed a vision-conditioned dynamics model that anticipates changes in system dynamics from visual context to inform a downstream MPC
- The approach was deployed in the autonomous racing stack on a physical Lexus LC 500 for racing in changing conditions on a test track

RWTH AACHEN UNIVERSITY | TEACHING & GRADUATE ASSISTANT

May 2019 – Mar 2022 | Aachen, Germany

- Assisted approx. 700 students in *Tutorial Control Engineering*
- Research on nonlinear model predictive control for fuel cell hybrid vehicles

SELECTED PUBLICATIONS

Full list: [website](#) / [Scholar](#)

Gibbs Variational Last Layers for Long-Horizon Dynamics Learning

P. Brunzema, H. Hose, S. Trimpe • *Neural Information Processing Systems (NeurIPS)*, 2026

Efficient Knowledge Transfer in Federated Bayesian Optimization

through Neural Network Surrogates

A. Gräfe, M. van Gemmeren, P. Brunzema, S. Trimpe • *Neural Information Processing Systems (NeurIPS)*, 2026

Agentic Bayesian Optimization through Surrogate-Augmented Autoresearch

P. Brunzema, L. C. Tiao, N. Le, K. De Angeli, Y. Xuan, D. Gligorijevic • *Under review*, 2026 — Internship at Meta

Vision-Conditioned Variational Bayesian Last Layer Dynamics Models

P. Brunzema, T. Lew, R. Zhang, T. Shirasawa, J. Subosits, M. Greiff • *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 — Internship at Toyota Research Institute

BayeSQP: Bayesian Optimization through Sequential Quadratic Programming

P. Brunzema, S. Trimpe • *Neural Information Processing Systems (NeurIPS)*, 2025 **Spotlight**

Bayesian Optimization via Continual Variational Last Layer Training

P. Brunzema, M. Jordahn, J. Wiles, S. Trimpe, J. Snoek, J. Harrison • *International Conference on Learning Representations (ICLR)*, 2025 **Spotlight** — Collab. with Google DeepMind, Vector Institute, and DTU

Event-Triggered Time-Varying Bayesian Optimization

P. Brunzema, A. von Rohr, F. Solowjow, S. Trimpe • *Transactions on Machine Learning Research (TMLR)*, 2025

On Controller Tuning with Time-Varying Bayesian Optimization

P. Brunzema, A. von Rohr, S. Trimpe • *IEEE Conference on Decision and Control (CDC)*, 2022

Reviewing for: AAAI, NeurIPS, ICLR, ECML, CDC, ACC, L4DC;

Journals: L-CSS, TCST, RA-L, Automatica